SIMATS ENGINEERING
ASSESSMENT TOOL

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Course: Theory of Computation (CSA13) | CO1: Analyse fundamental mathematical concepts of automata theory for formal languages and decision problems.

Question 1

Consider the NFA given (over $\Sigma = \{a, b\}$) which accepts some language L. Construct a DFA that represents the same language L.

Step 1: NFA transition table (from the given diagram)

Start state: q0 Final state: q2

State	a	b
q0	{q1}	{q2}
q1	{q0}	{q1}
q2	∅	{q0, q1}

Step 2: Subset construction (each DFA state = union of NFA states)

DFA State	NFA Subset	a	b	Final?
A	{q0} (start)	B	C	-
B	{q1}	A	B	-
C	{q2}	D	E	-
D	∅ (trap)	D	D	-
E	{q0, q1}	E	F	-
F	{q1, q2}	A	E	Yes

Result: DFA start state = A. Final states = {C, F} (every subset containing q2 is a final state).

Question 2

Let L be the language of words of length at least 2 who's last two letters are different from each other (over $\Sigma = \{a, b\}$). Example: abaab ∈ L, but bb ∉ L. Construct an NFA for L, then draw the equivalent DFA.

Step 1: NFA design idea

Stay in the start state q0 for any prefix (self-loop on a and b). Nondeterministically "guess" that the current symbol is the second-last symbol, move to q1 (if it was 'a') or q2 (if it was 'b'), then check the next symbol is different and move to the final state qf.

NFA transition table

State	a	b
q0 (start)	{q0, q1}	{q0, q2}
q1	∅	{qf}
q2	{qf}	∅
qf (final)	∅	∅

Step 2: Equivalent DFA (subset construction)

DFA State	NFA Subset	a	b	Final?

A	{q0} (start)	B	C	-
B	{q0, q1}	B	D	-
C	{q0, q2}	E	C	-
D	{q0, q2, qf}	E	C	Yes
E	{q0, q1, qf}	B	D	Yes

Result: DFA start state = A. Final states = {D, E}.

Question 3

Design an NFA that takes a binary string as input and accepts the string only if the last two digits are the same (00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA.

Formal definition of the NFA

N = (Q, Σ, δ, q0, F)

Q = {q0, q1, q2, qf}   Σ = {0, 1}   Start state = q0   F = {qf}

State	0	1
q0 (start)	{q0, q1}	{q0, q2}
q1	{qf}	∅
q2	∅	{qf}
qf (final)	∅	∅

Verifying that 01011 is accepted

Since the string ends in 11, we take the branch that guesses the second-last '1' is the start of the matching pair:

Symbol read	0	1	0	1	1
State reached	q0	q0	q0	q2	qf

Path: q0 →(0) q0 →(1) q0 →(0) q0 →(1) q2 →(1) qf. The string is fully consumed and the NFA ends in the final state qf, so 01011 is accepted.

Question 4

Consider the NFA given which accepts the language representing the set of all strings having 001 as a substring, over Σ = {0, 1}. Construct a DFA that represents the same language.

Step 1: NFA transition table

Start state: q0   Final state: q3

State	0	1
q0	{q0, q1}	{q0}
q1	{q2}	∅
q2	{q2}	{q3}
q3	{q3}	{q3}

Step 2: Subset construction

DFA State	NFA Subset	0	1	Final?
A	{q0} (start)	B	A	-
B	{q0, q1}	C	A	-
C	{q0, q1, q2}	C	D	-
D	{q0, q3}	E	D	Yes
E	{q0, q1, q3}	F	D	Yes
F	{q0, q1, q2, q3}	F	D	Yes

Result: DFA start state = A. Final states = {D, E, F}.

Question 5

A NFA with ε-transitions is given. Find ε-closure for each state and design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states.

Given ε-NFA (from the diagram)

q0 (start): self-loop on 1, ε → q1 | q1: self-loop on 0, ε → q2 | q2 (final): self-loop on 0 and 1

Step 1: ε-closures

State	ε-closure
q0	{q0, q1, q2}
q1	{q1, q2}
q2	{q2}

Step 2: Subset construction using ε-closures

DFA State	NFA Subset	0	1	Final?
A	{q0, q1, q2} (start)	B	A	Yes
B	{q1, q2}	B	C	Yes
C	{q2}	C	C	Yes

Result: DFA start state = A. Final states = {A, B, C}. Every reachable DFA state contains q2, which is the accepting state of the ε-NFA. Therefore, all reachable DFA states are accepting (final) states.